

CY 301 : Principles and Theories of Physical Chemistry

Course Code : CY 301

Course Name : Principles and Theories of Physical Chemistry

L-T-P-C : 3-0-0-3

Intended for : BS Chemical Sciences

Prerequisite : None

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

- **Gaseous State and Fluids:** Gas laws, distribution of molecular speeds, kinetic energy distribution, molar heat capacity of gases, virial expressions, collision of gas molecules and mean free path. Viscosity of fluids, viscosity coefficient, temperature dependence of viscosity, surface tension of liquids, capillary rise, measurement of surface tension, temperature dependence of surface tension. (12 Hours)
- **Concepts of Thermodynamics:** Equilibrium and concept of temperature, the zeroth-law of thermodynamics, first law of thermodynamics, state and path functions, extensive and intensive properties, equation of state, work, heat, internal energy, heat capacity and concept of enthalpy, second law of thermodynamics, reversible and irreversible process, heat engines, Carnot cycle, concept of entropy, free energy, criteria for equilibrium and stability, third law of thermodynamics, concept of the absolute zero temperature and Nernst heat theorem. (12 Hours)
- **Chemical Kinetics and Photochemistry:** Rate laws and rate constants, order and molecularity of reactions, determination of order, kinetics of zero-, first- and second-order reactions, parallel, reversible and consecutive reactions, rate-determining and steady-state approximation, temperature dependence of rate constant, potential energy surface, Frank-Condon principle, decay of excited states, fluorescence and phosphorescence, Jablonsky diagram, laws of photochemistry, quantum yield. (12 Hours)
- **Conductance and Electrochemistry:** Mechanism of electrolysis and Faraday's law, strong and weak electrolytes, conductance, electrolytic conductance, ionic conductance, conductometric titration, estimation of solubility product. Types of electrochemical cells, cell reactions, EMF and change thermodynamics properties, Nernst equation, standard cells, half-cells/electrodes. (6 Hours)

Textbooks:

1. Peter Atkins, Julio de Paula, James Keeler, **Physical Chemistry**, Oxford University Press, 2018.
2. Donald A. McQuarrie & John D. Simons, **Physical Chemistry: A molecular approach**, Indian Reprint, Viva books, 2019.

References:

1. Ira N. Levine, **Physical Chemistry**, McGraw Hill Book Co., 2008.
2. G. W. Castellan, **Physical Chemistry**, Narosa Publications, 2004.